Sai Abhinav Pattoor

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SUMMARY

Data Engineer–focused Computer Science senior (May 2027, 3.95 GPA) who builds and runs Python ETL and data-quality pipelines in production: 50,000+ records reconciled weekly and 90% of duplicate student records resolved automatically. Also builds AI tooling (RAG and MCP on the Claude API) and full-stack apps. Seeking new-grad Data Engineer and Software Engineer roles.

EDUCATION

Texas Tech University Expected May 2027
B.S. Computer Science, Minor in Mathematics | GPA 3.95/4.00 | President's Honor List: Fall 2025, Spring 2026
Coursework: Data Structures, Algorithms, Databases, Operating Systems

EXPERIENCE

Texas Tech University, Enrollment Management IT Lubbock, TX
Student Assistant, Data & Systems Engineering Sep 2025 – Present

- Built a Python duplicate-resolution tool that automatically resolves 90% of duplicate student records using match and survivorship rules; ambiguous cases are routed to human review, and I work through thousands of them weekly.
- Reconcile 50,000+ records weekly across Informatica ETL loads (up to 10,000 records each), checking every load against source files with SOQL to catch rejected and missing records.
- Wrote Python pre- and post-processing for 3+ recurring vendor-file import jobs that clean and validate data before Informatica loads, saving 5+ hours of manual work per week; 15 automated tests cover the scripts, Informatica jobs, and merge logic.
- Develop and test Salesforce Apex and Flow solutions through QA, UAT, and production; built recruitment dashboards for application tracking.

IT Systems Intern, Automation & Data Quality Jun – Aug 2025

- Replaced hard-coded routing logic with a configuration-driven lookup covering 100+ academic programs, so non-technical staff can update mappings without code changes.
- Fixed historical reporting by replacing a calculated field with an automated trigger that stores values permanently, and enforced unique record IDs to stop duplicate-record import failures.
- Tested UI components for accessibility (WCAG 2.2 Resize Text and Reflow) at 200% and 400% zoom.

ProcureX Technologies Hyderabad, India
Technical Systems & Data Engineering Intern May – Aug 2024

- Validated 40+ data flows across 8 components of a procurement platform using Python and Excel, catching 15 ID and unit mismatches before integration and documenting each for engineering review.

PROJECTS

Austin Real Estate Analyzer | Python, pandas, scikit-learn, SQL, SQLite, Streamlit, AWS, Airflow Dec 2025 – Nov 2026

- Built an end-to-end ETL pipeline that ingests 7+ data sources, engineers features, and applies K-Means clustering and scoring to rank 50+ Austin ZIP codes in a filterable Streamlit app.
- Designed a SQLite layer with idempotent upserts, dated snapshots, and import audit logs so reruns are safe; 14 automated tests.
- Deployed the pipeline on AWS and orchestrated recurring runs with Airflow.

ZipRAG: Austin Neighborhood Q&A (RAG + MCP) | Python, Claude API, OpenAI, MCP, SQLite, Streamlit Aug 2026

- Built a retrieval-augmented assistant over Austin neighborhood documents and the analyzer's ZIP scores; Claude chooses between document search and score lookup through tool use and cites its sources.
- Exposed both tools as an MCP server so Claude Desktop can call them directly; wrote a 10-question evaluation set to test whether the right source lands in the top 3 retrieved results (hit@3).

The Prospector | Node.js, Express.js, MongoDB | Runner-up, HackWesTX Sep 2025

- Co-developed a full-stack search app that merges 5 external APIs into one deduplicated result set, with authentication, favorites, and search history.

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript, Apex, SOQL
Data: ETL, Informatica, pandas, scikit-learn, SQLite, MongoDB, data validation, reconciliation, deduplication
Backend & Cloud: Node.js, Express.js, REST APIs, Streamlit, Salesforce (Flow, Lightning), AWS, Airflow
AI: RAG, MCP, Claude API, OpenAI API
Practices: Git/GitHub, automated testing, UAT, accessibility testing
Certifications (2026): Claude API, Model Context Protocol, Claude in Amazon Bedrock (Anthropic); Agents and Workflows, RAG (OpenAI)